

What is Cryptography

- Cryptography is the science of securing information by transforming it into an unreadable format, ensuring only authorized parties can access it.
- It protects Confidentiality, Integrity, Authentication, and Non-repudiation of data.
- Used in secure communication, online transactions, digital identities, and cybersecurity.

Core Concepts of Cryptography

Encryption

- The process of converting plain text into ciphertext using an algorithm and a key.
- Ensures confidentiality
- Symmetric Encryption → Same key is used for both encryption & decryption (e.g., AES, DES).
- Asymmetric Encryption → Different keys: Public Key (encrypt) and Private Key (decrypt) (e.g., RSA, ECC).
- Messaging apps (WhatsApp, Signal) use encryption to protect chats.

Hashing

- A one-way function that converts input data into a fixed-length string.
- Ensures integrity & Irreversible
- Even a small input change produces a completely different hash (avalanche effect).
- Passwords are stored as hashes instead of plain text in databases.

Digital Signatures

- A cryptographic mechanism that verifies the authenticity and integrity of digital data.
- Sender hashes the message and encrypts the hash with their private key → creating a digital signature.
- Receiver decrypts the signature using the sender's public key and verifies the hash.
- Authentication & Integrity
- Provides non-repudiation (sender cannot deny sending).
- Used in software distribution, SSL certificates, blockchain transactions.

Public Key & Private Key

- A pair of mathematically related cryptographic keys used in asymmetric encryption.
- Public Key: Shared openly; used to encrypt data or verify a digital signature.
- Private Key: Kept secret; used to decrypt data or create a digital signature.
- Data encrypted with a public key can only be decrypted with the corresponding private key.
- Ensures secure communication without exchanging secret keys.

Why Cryptography is Important in Cyber Security?

- Confidentiality
- Integrity
- Authentication
- Non-repudiation
- Real-Life Applications: Online banking, e-commerce, VPNs, cloud storage, digital identity verification.